

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 59. For Loop

Section 1: Lecture Summary

This lecture introduces the **for loop** in Python, one of the most common looping constructs used for iteration over sequences. The lecture first explains the **range() function/object**, which is often used with for loops to generate a sequence of numbers. The range function takes up to three parameters: start, stop (required), and step, with default values for start (0) and step (1). Various examples demonstrate how to use range with different parameters, including negative start and step to generate decreasing sequences.

After covering the range function, the lecture explains the syntax and working of the for loop: `for <variable> in <sequence>: <statements>`. The loop iterates over elements in the sequence (like a range object or string), assigning each value to a variable for use inside the loop body. The lecture further illustrates how for loops iterate over strings character-by-character.

A flowchart representation of the for loop is presented to clarify its operation — repeatedly obtaining the next element in the sequence and executing the body until no elements remain.

Finally, the lecture includes practical demonstrations of for loops in VS Code, showing basic loops with range and string iteration, validated by debugging sessions stepping through each iteration.

Section 2: Key Concepts and Explanations

- Loops and For Loop Concept

- Loop means repetition of statements.
- For loop specifically iterates over a sequence of values, often generated by `range()`.
- For loop is also called a "for each" loop because it processes each item in a sequence one by one.

- `range()` Function

- Syntax: ``range(start, stop, step)``
- Parameters:
 - ``start``: Optional, default = 0.
 - ``stop``: Required, loop runs up to but excludes this value.
 - ``step``: Optional, default = 1, which is the increment.
- Examples:
 - ``range(5)`` → yields 0,1,2,3,4
 - ``range(1,5)`` → yields 1,2,3,4
 - ``range(6,11)`` → yields 6,7,8,9,10
 - ``range(0,10,2)`` → yields 0,2,4,6,8
- Negative values and steps allowed:
 - ``range(-5,-1)`` → yields -5,-4,-3,-2
 - ``range(10,5,-1)`` → yields 10,9,8,7,6 (counts down)

- For Loop Syntax

- General form:

```
for <variable> in <sequence>:  
    <statements>
```

- ``<variable>`` takes each element of the sequence one by one.

- ``<sequence>`` can be any iterable, such as a range object or string.
- The loop continues until all elements are exhausted.

- For Loop with Strings

- Strings are sequences; for loop iterates over each character.
- Example: iterating over `"python"` will assign 'p', then 'y', then 't', etc. to the loop variable.

- Flow of for loop

- Check if sequence has next element.
- Assign the next element to the loop variable.
- Execute the loop body statements.
- Repeat until no more elements, then continue after the loop.

Section 3: Example Code and Use Cases

Example 1: Using range with one argument

```
for i in range(5):  
    print(i)
```

Prints numbers 0 to 4

Example 2: Using range with start and stop

```
for i in range(1, 5):  
    print(i)
```

Prints numbers 1 to 4

Example 3: Using range with start, stop and step

```
for i in range(0, 10, 2):  
    print(i)
```

Prints even numbers 0, 2, 4, 6, 8

Example 4: Negative start and stop values

```
for i in range(-5, -1):  
    print(i)
```

Prints -5, -4, -3, -2

Example 5: Decrementing range with negative step

```
for i in range(10, 5, -1):  
    print(i)
```

Prints 10, 9, 8, 7, 6

Example 6: For loop iterating over a string

```
s1 = "python"  
for x in s1:  
    print(x)
```

Prints each character: p, y, t, h, o, n

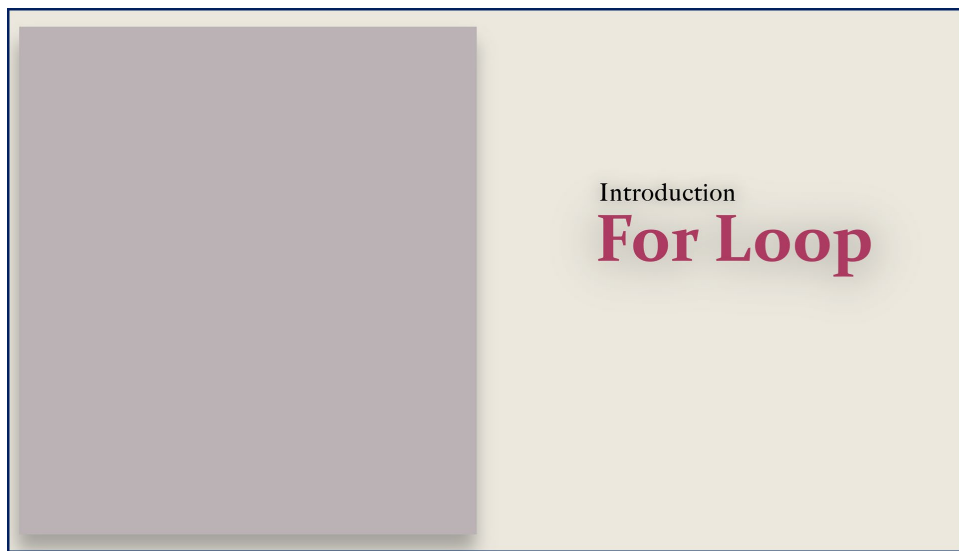
Section 4: Key Takeaways

- The **for loop** is used to iterate over sequences (like those produced by `range()` or strings).
- The **range() function** is fundamental to creating number sequences for loops, with flexible start, stop, and step parameters.
- The `'stop'` parameter in range is exclusive; the sequence stops before this value.
- Both **start** and **step** have default values (0 and 1 respectively) if not provided.

- For loops process each item one at a time in order, not all at once.
- Strings are sequences too, so for loops can iterate through each character in a string.
- Flowchart logic for for loops: repeatedly get next item in sequence → execute loop body → continue until sequence exhausted.
- Practical debugging helps see the for loop variable changing and statements executing per iteration.
- For loops in Python differ from some other languages by directly iterating over values in sequences (for-each style), not just index-based iteration.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



For Loop

Introduction

For Loop

Introduction

- `range( )` function

For Loop

Introduction

- range() function
- For Loop

For Loop

Introduction

- range() function
- For each Loop

- `range()` function

`range()`
function

range()
function

range ()

range()
function

range (start)

range()
function

range (start , stop)

range()
function

range (start , stop , step)

range()
function

range (start , stop, step)

Optional

range()
function

range (0 , stop, step)

Optional

range()
function

range (start , stop, step)

Required

range()
function

range (start , stop , step)

Optional

range()
function

range (start , stop , 1)

Optional



A diagram showing a light beige rectangular area divided into two parts. The left part is a solid grey rectangle. The right part contains the text 'range (start , stop, step)' in a monospace font. The word 'range' is green, and the parameters 'start', 'stop', and 'step' are purple.

range (start , stop, step)



A diagram showing a light beige rectangular area divided into two parts. The left part is a solid grey rectangle. The right part contains the text 'range (start , stop, step)' in a monospace font. The word 'range' is black, and the parameters 'start', 'stop', and 'step' are purple.

range (start , stop, step)

range (start , stop, step)

range(5)

range (start , stop, step)

range(5)

0	1	2	3	4
---	---	---	---	---

range (start , stop, step)

range(5)

0	1	2	3	4
---	---	---	---	---

range(, 5)

range (start , stop, step)

range(5)

0	1	2	3	4
---	---	---	---	---

range(1, 5)

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6,)

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6, 11)

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6, 11)

6

7

8

9

10

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6, 11)

6

7

8

9

10

range(0, 10,)

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6, 11)

6

7

8

9

10

range(0, 10, 2)

range (start , stop, step)

range(5)

0

1

2

3

4

range(1, 5)

1

2

3

4

range(6, 11)

6

7

8

9

10

range(0, 10, 2)

0

2

4

6

8

range (start , stop, step)

range(-5)

range (start , stop, step)

range(-5 , -1)

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2,)

range (start , stop, step)

range(-5 , -1)

-5	-4	-3	-2
----	----	----	----

range(-2, 3)

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2, 3)

-2

-1

0

1

2

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2, 3)

-2

-1

0

1

2

range(10, ,)

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2, 3)

-2

-1

0

1

2

range(10, 5,)

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2, 3)

-2

-1

0

1

2

range(10, 5, -1)

range (start , stop, step)

range(-5 , -1)

-5

-4

-3

-2

range(-2, 3)

-2

-1

0

1

2

range(10, 5, -1)

10

9

8

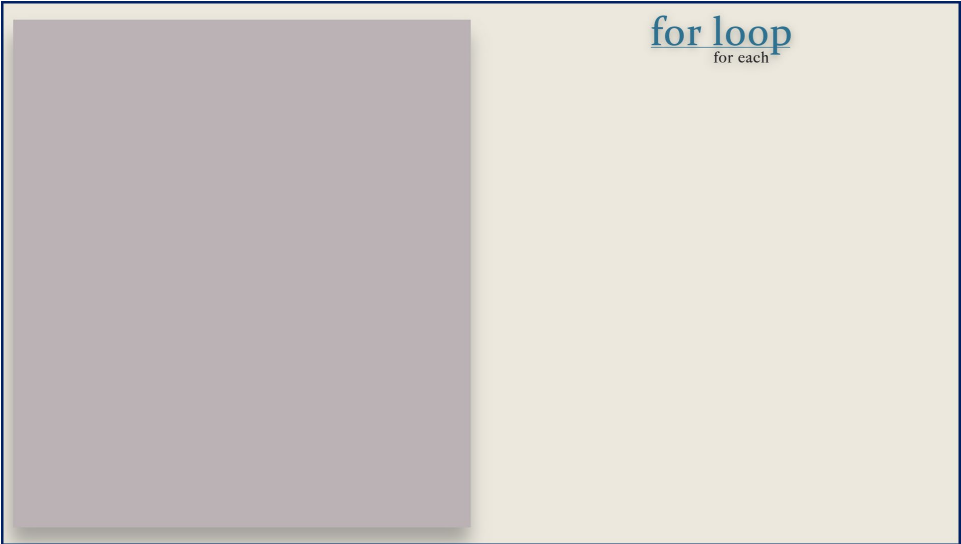
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6



A diagram showing a light beige rectangle with a dark blue border. Inside, on the left, is a smaller, semi-transparent grey rectangle. To the right of the grey rectangle, the text "• for loop" is written in dark blue, with "for each" in a smaller font below it.

- for loop
for each



A diagram showing a light beige rectangle with a dark blue border. Inside, on the left, is a smaller, semi-transparent grey rectangle. To the right of the grey rectangle, the text "for loop" is written in dark blue, with "for each" in a smaller font below it.

for loop
for each

for loop
for each

Syntax

for loop
for each

Syntax

Working

for loop

for each

Syntax

Working

Flow Chart

for loop

for each

```
for <var> in <sequence>:
```

```
    Statement 1
```

```
    Statement 2
```

for loop

for each

```
for i in <sequence>:
```

```
    Statement 1
```

```
    Statement 2
```

for loop
for each

```
for i in range(5):
```

```
    Statement 1
```

```
    Statement 2
```

for loop
for each

```
for i in range(5):
```

```
    Statement 1
```



for loop
for each

```
for i in range(5):  
    print(i)
```



for loop
for each

```
for i in range(5):  
    print(i)
```





for loop
for each

→ **for** **i** **in** **range(5):**
print(i)



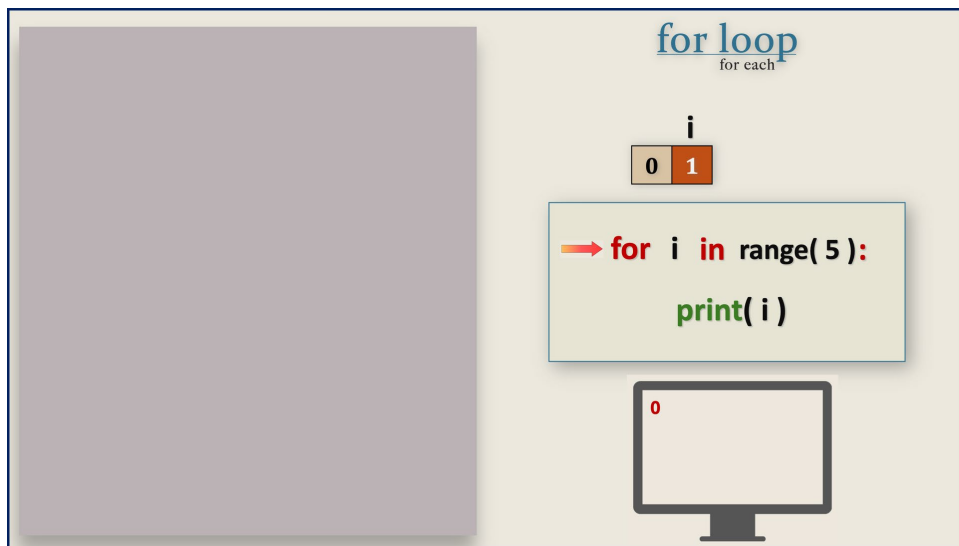
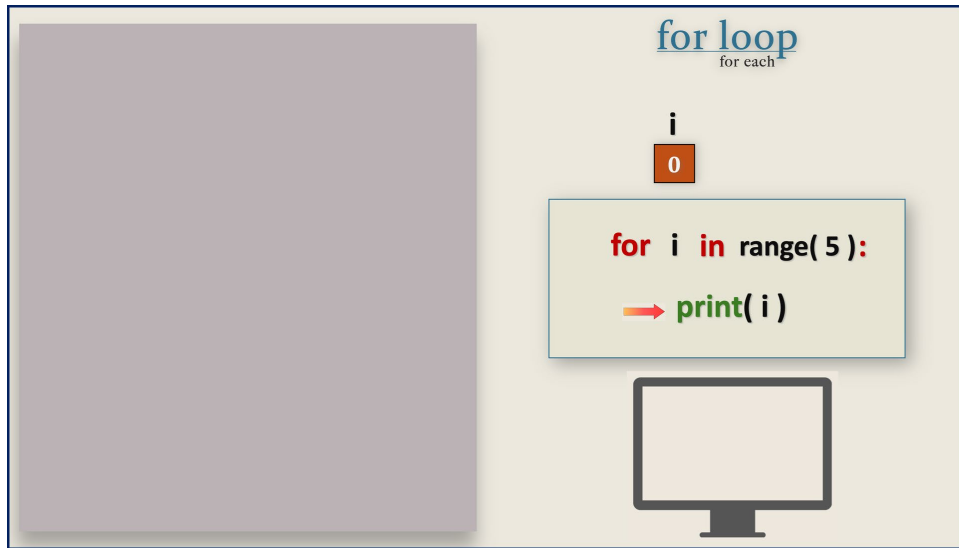


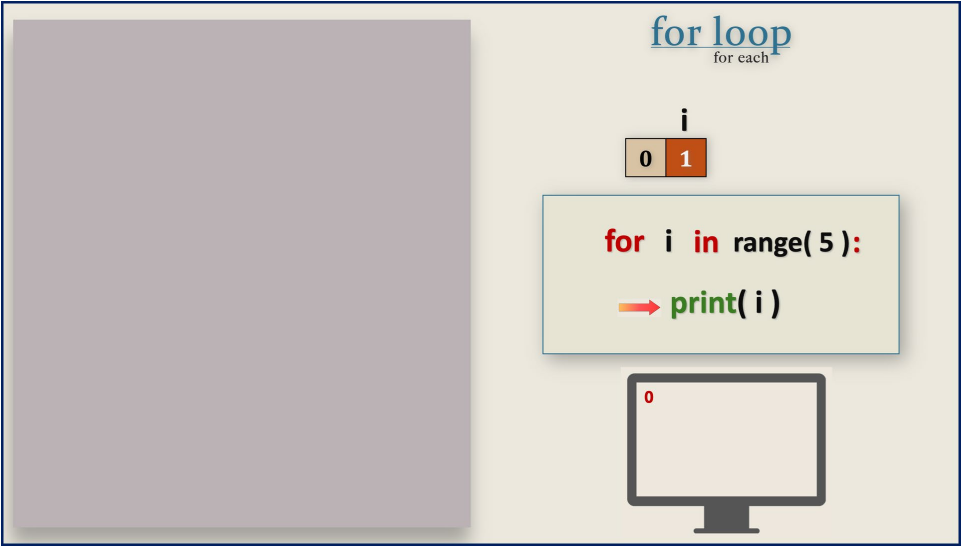
for loop
for each

i
0

→ **for** **i** **in** **range(5):**
print(i)







The diagram illustrates a Python for loop. On the left is a large, empty grey rectangle. To its right, the text "for loop" is displayed in blue, with "for each" in smaller black text below it. Below this, a variable `i` is shown above a row of two boxes: a light tan box containing the number `0` and a dark orange box containing the number `1`. Further down, a light yellow box contains the code `for i in range(5):` in red and `→ print(i)` in green. At the bottom right, a computer monitor icon displays the number `0` in red.



for loop
for each

i

0	1	2
---	---	---

→ **for i in range(5):**

print(i)

0
1



for loop
for each

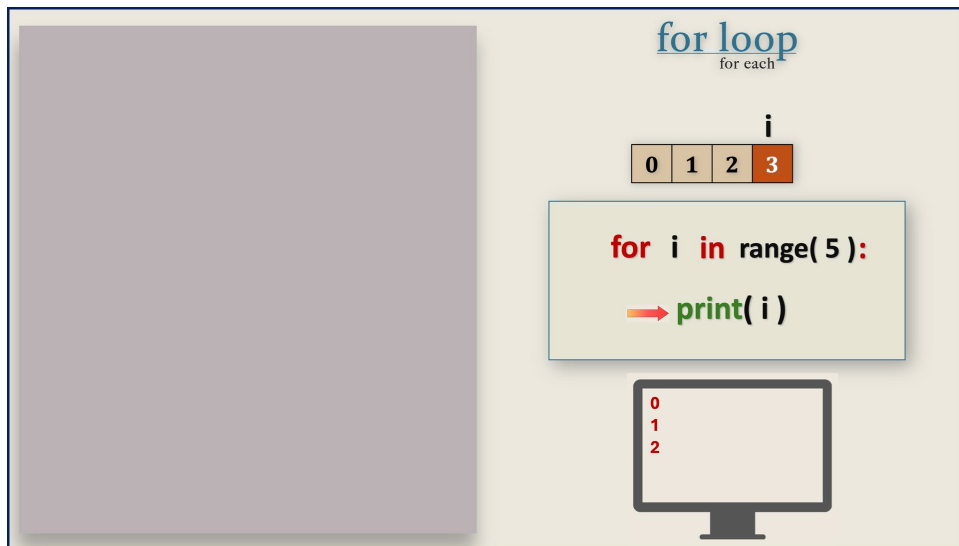
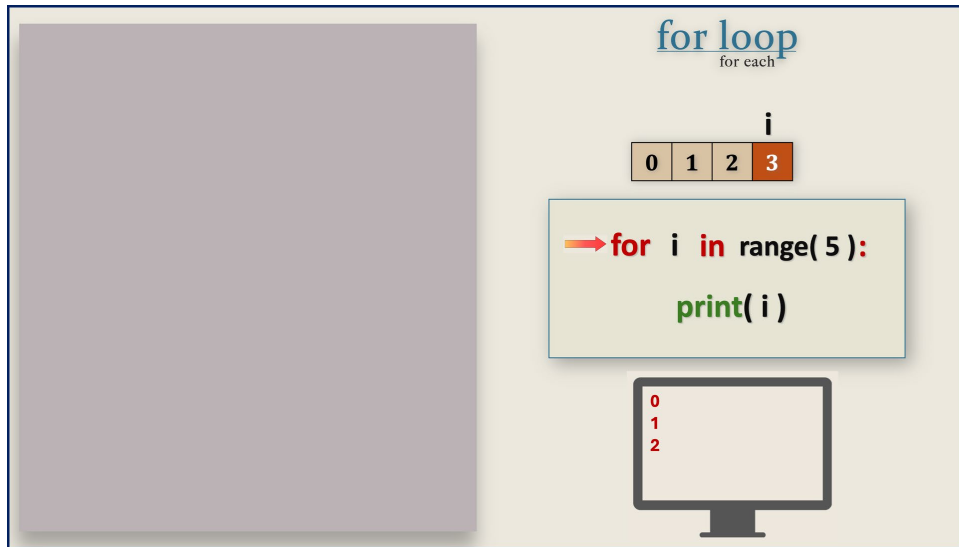
i

0	1	2
---	---	---

for i in range(5):

→ **print(i)**

0
1



for loop

for each

i

01234

→ for i in range(5):

print(i)

0

1

2

3

for loop

for each

i

01234

for i in range(5):

→ print(i)

0

1

2

3

for loop
for each

i

0	1	2	3	4
---	---	---	---	---

→ for i in range(5):

print(i)

0
1
2
3
4

for loop
for each

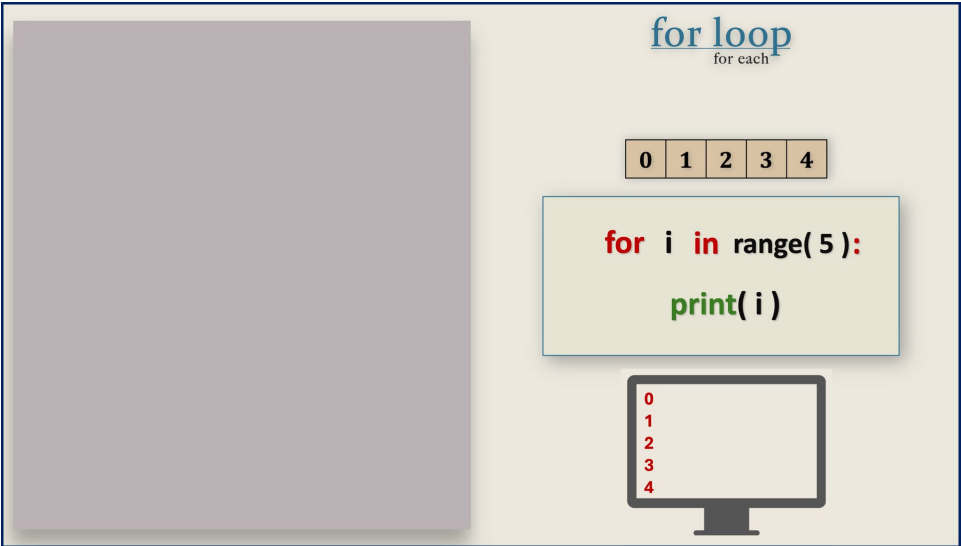
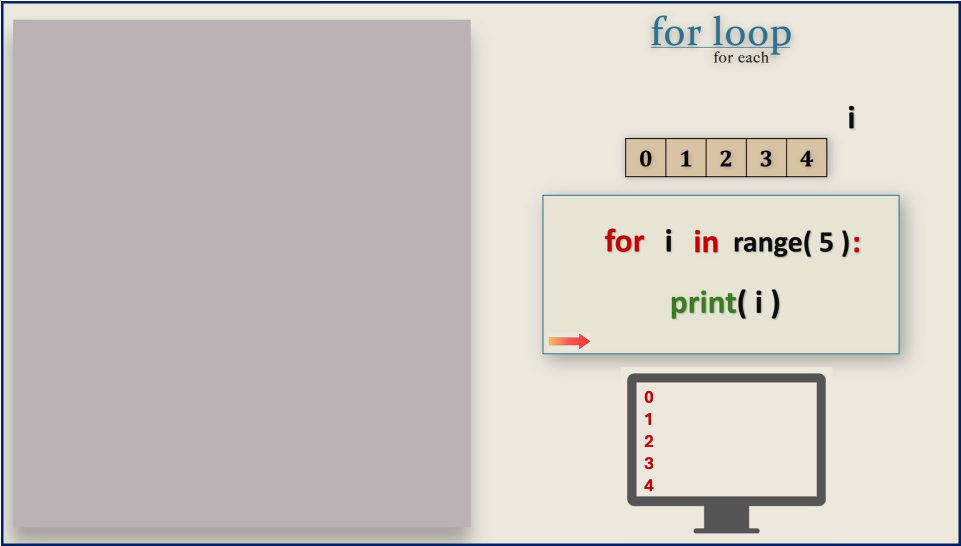
i

0	1	2	3	4
---	---	---	---	---

→ for i in range(5):

print(i)

0
1
2
3
4



for loop
for each

Working with String

for loop
for each

```
s1 = 'python'
```

for loop
for each

```
s1 = 'python'
```

```
for x in s1 :
```

for loop

for each

```
s1 = 'python'  
for x in s1 :  
    print( x )
```

for loop

for each

```
s1 = 'python'  
for x in s1 :  
    print( x )
```

for loop

for each

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1

p

y

t

h

o

n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1 **x**
p y t h o n

```
s1 = 'python'
```

```
for x in s1 :
```

```
    print( x )
```

for loop

for each

x

s1p y t h o n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

x

s1p y t h o n

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)

for loop

for each

s1

p

y

t

h

o

n

x

s1 = 'python'

for x in s1 :

print(x)



for loop
for each

s1

p	y	t	h	o	n
---	---	---	---	---	---

```
s1 = 'python'  
for x in s1 :  
    print(x)
```



for loop
for each

Flow Chart

for loop

Flow Chart

```
for <var> in <sequence>:  
    Statement 1
```

for loop

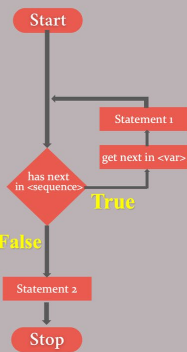
Flow Chart

```
for <var> in <sequence>:
```

```
    Statement 1
```

```
    Statement 2
```

Flow Chart



for loop

Flow Chart

```
for <var> in <sequence>:
```

```
    Statement 1
```

```
    Statement 2
```